OMNI CAMERA ROVER | REV E PROTOTYPE

1. Controller

File: controller.kicad_sch

2. Motors

File: motors.kicad_sch

3. Charging

File: charging.kicad_sch

Battery protection

File: battery_protection.kicad_sch

5. Interlock

File: interlock.kicad_sch

4. Regulator

File: regulator.kicad_sch

Flex camera

File: camera.kicad_sch

Cell undervoltage load shutdown

File: cell_undervoltage.kicad_sch

Cell temperature charging window

File: charge_temperature.kicad_sch

Rev F USB and drive power path

File: discharge_cutoff.kicad_sch

Editable functional sheets. Global net names link sheets.
Prototype: electrical, thermal and mechanical qualification remains open.

PWR_FLAG

U2_VINT

PWR_FLAG

GND

PWR_FLAG

VMOTOR

PWR_FLAG

USB_5V

PWR_FLAG

U3_VINT

PWR_FLAG

CELL_P

PWR_FLAG

PROT_VSS_SENSE

Nathaniel Sanford

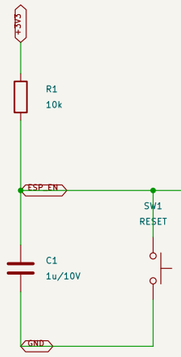
Sheet: /
File: omni_ugv.kicad_sch

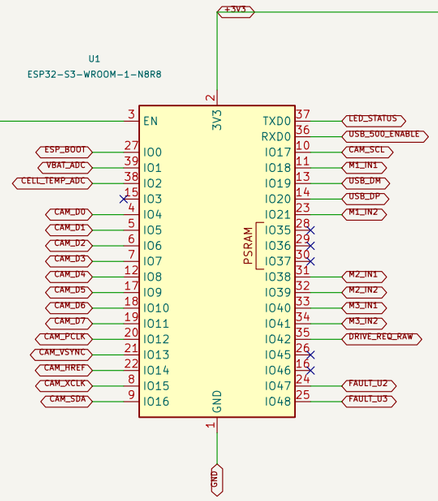
Title: Omni UGV / Rev E

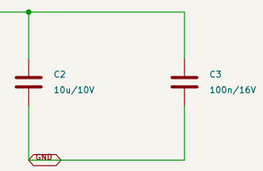
Size: A3	Date: 2026-11-07	**Rev: F — integration draft**
KiCad E.D.A. 10.0.8		Id: 1/11

Controller, camera and service points

RESET / enable RC







Waveshare OV2640 | documented 2 x 9 interface

Status indicator



BOOT strap



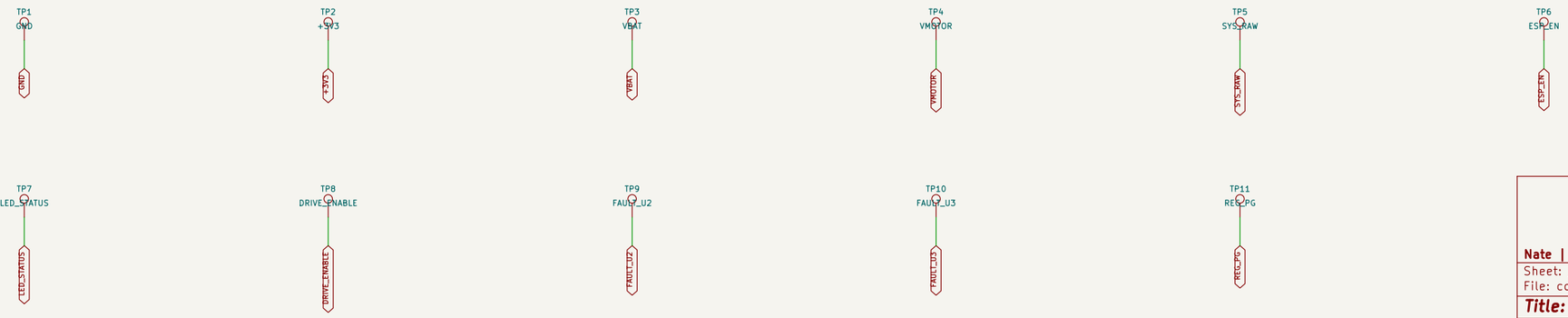
Camera reset RC





[SPEC] GPIO35 / 36 / 37 reserved for N8R8 PSRAM.

Service points (named nets continue on other sheets)



Nate | Omni camera rover

Sheet: /1. Controller/
File: controller.kicad_sch

Title: Controller, camera and service points

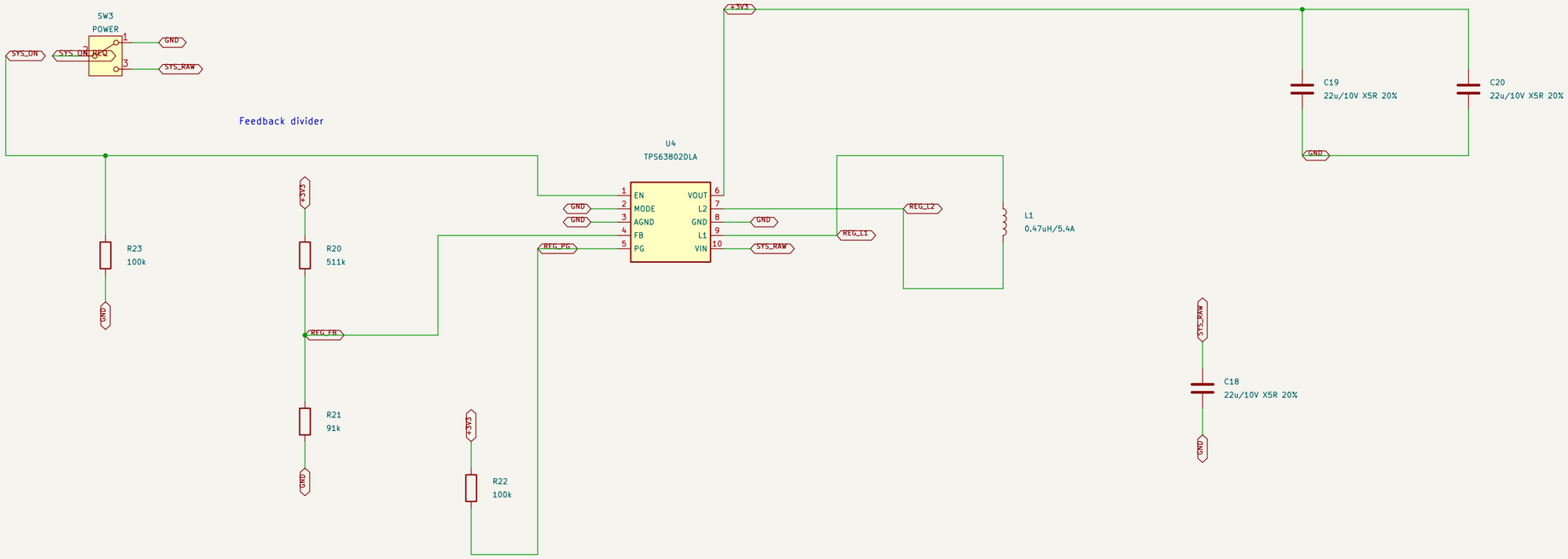
Size: A3	Date:	**Rev: E – prototype**
KiCad E.D.A. 10.0.8		Id: 2/11

3.3 V buck–boost supply

System enable

Feedback divider

Output capacitors

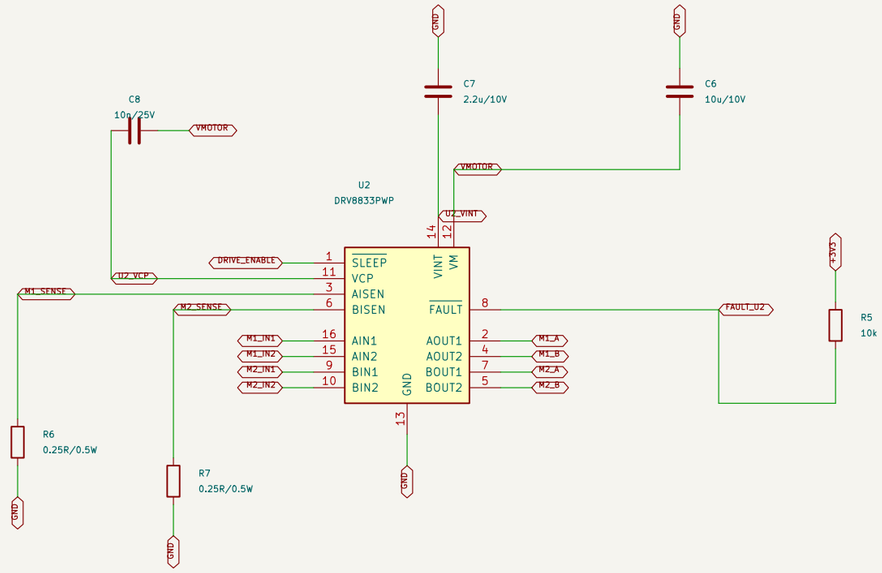


[TARGET] Keep switching loops short. Place input/output capacitors at U4.
[TBD–MEASURE] Verify startup, load transients and capacitor effective capacitance.

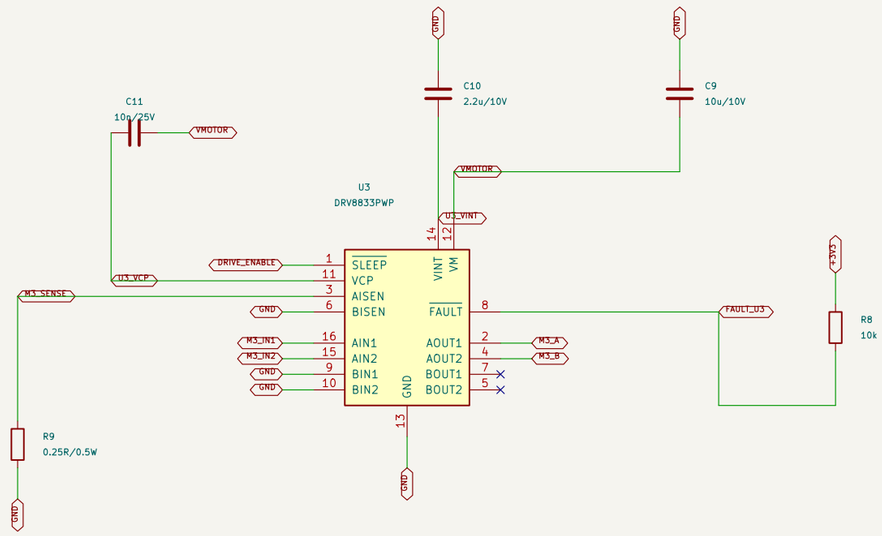
Nate | Omni camera rover		
Sheet: /4. Regulator/		
File: regulator.kicad_sch		
Title: 3.3 V buck–boost supply		
Size: A3	Date:	**Rev: E – prototype**
KiCad E.D.A. 10.0.8		Id: 3/11

Three independent motor channels

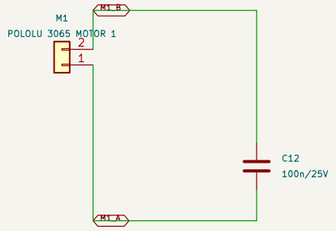
U2: wheels 1 and 2

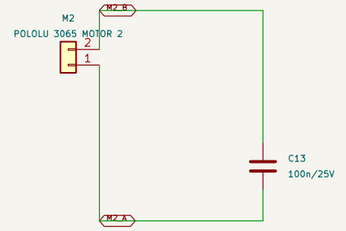


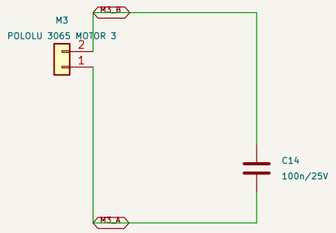
U3: wheel 3; unused bridge pins retained as designed

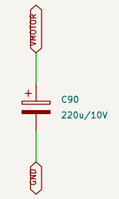


Motor terminal pads and suppression capacitors







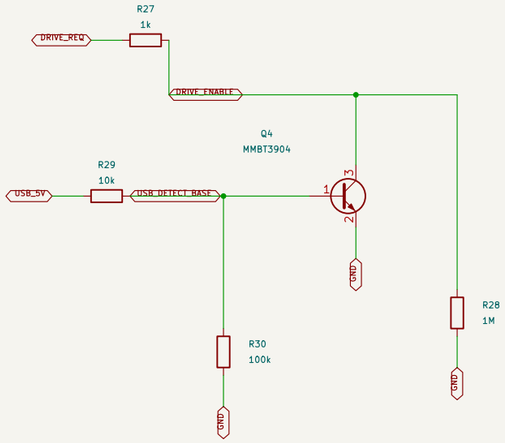


[TARGET] Route motor current away from logic returns.
[TBD-MEASURE] Braking overshoot, driver temperature and carpet load.

Nate | Omni camera rover	
Sheet: /2. Motors/	
File: motors.kicad_sch	
Title: Three independent motor channels	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: E — prototype**
	Id: 4/11

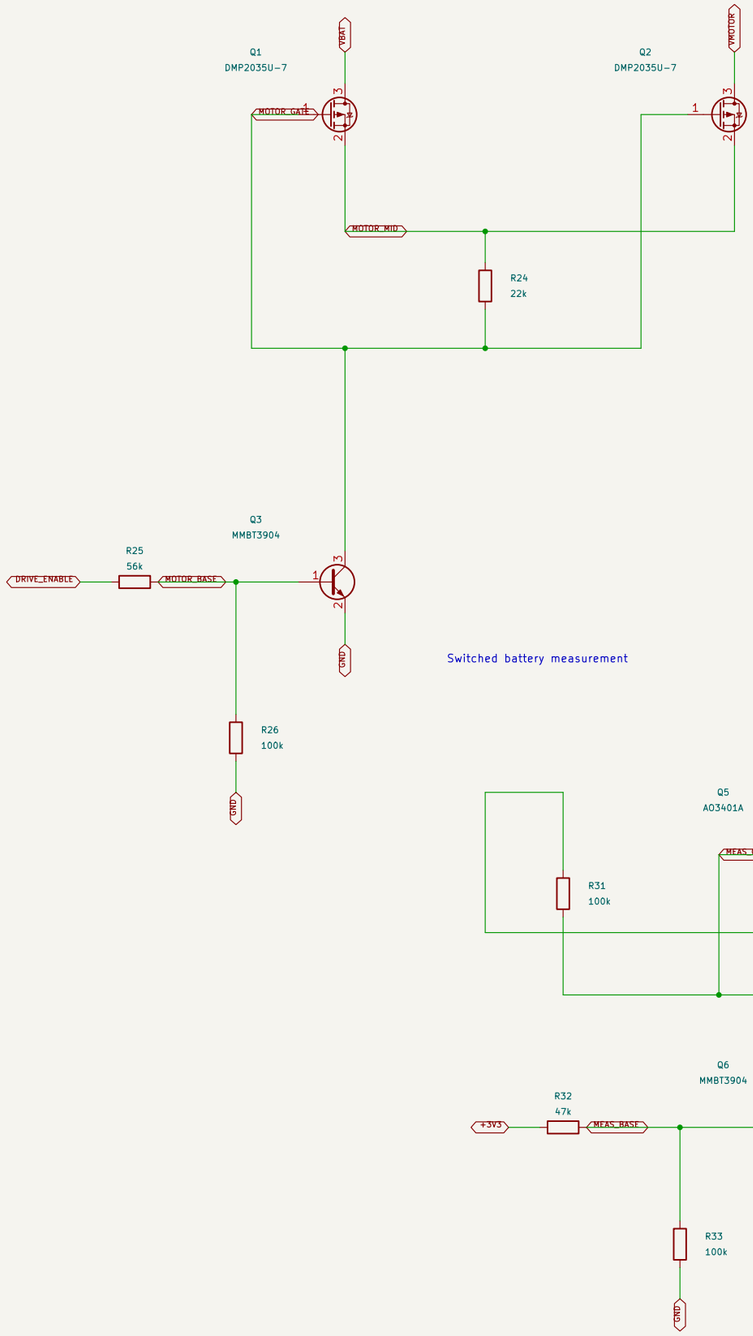
USB motor interlock and battery measurement

USB presence clamps DRIVE_ENABLE

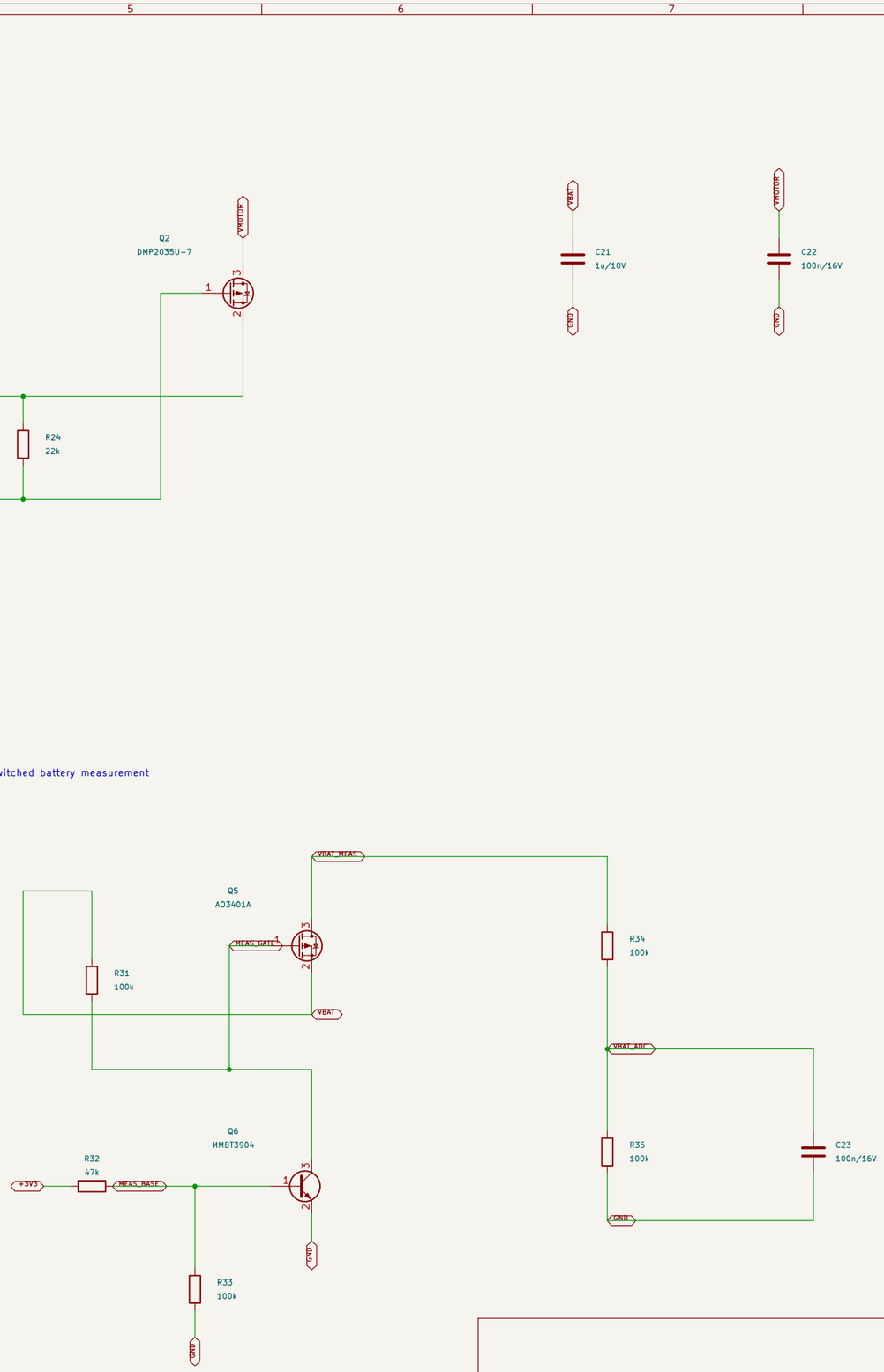


[TARGET] Reset / absent commands keep motors disabled.
[TBD-MEASURE] Verify USB insertion/removal and motor braking transients.

Back-to-back motor supply switch



Switched battery measurement



Nate | Omni camera rover	
Sheet: /5. Interlock/	
File: interlock.kicad_sch	
Title: USB motor interlock and battery measurement	
Size: A3 | Date:	**Rev: E – prototype**
KiCad E.D.A. 10.0.8	Id: 5/11

USB input, charging and cell connections

USB data to ESP32 native USB

J1
USB-C

R10
5.1k

R11
5.1k

U6
USBLC6-2SC6

R12
22R

R13
22R

Raw cell -> protection sheet

J2
RAW 1S CELL 4.2V MAX

Charge / system power path

U5
BQ24074RGT

C15
1u/10V

C17
10u/10V

C16
10u/10V

R16
46.4k

R17
100k

R14
3.57k

R15
3.24k

R19
100k

R18
2.2k

D2
CHARGE AMBER

Cell-bonded charger probe

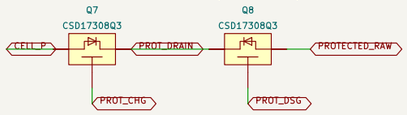
J4
CELL NTC - CHARGER

[SPEC] J2 CELL_N is not system GND: R36 is the only intended return path.
J4: independent insulated two-wire 103AT-2 probe on the cell.
Motor current uses protected VBAT, not charger SYS_RAW output.

Nate | Omni camera rover	
Sheet: /3. Charging/	
File: charging.kicad_sch	
Title: USB input, charging and cell connections	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: E - prototype**
	Id: 6/11

1. Cell and protection power path

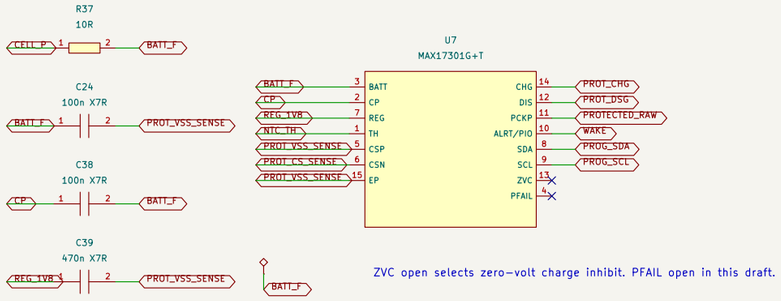
Q1 source faces cell. Q2 source faces protected output. Drains are common.





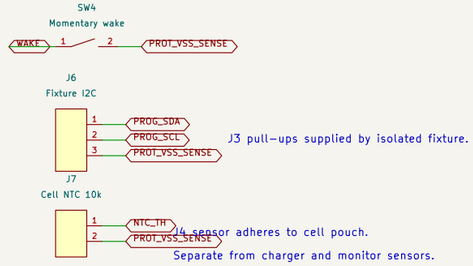
R1 carries all pack return current. CSP_K and CSN_K use separate Kelvin pads.

2. MAX17301G+T and local supply



ZVC open selects zero-volt charge inhibit. PFAIL open in this draft.

3. Wake, provisioning and cell probe

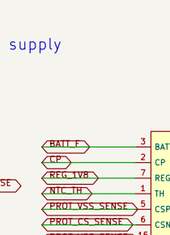


J3 pull-ups supplied by isolated fixture.

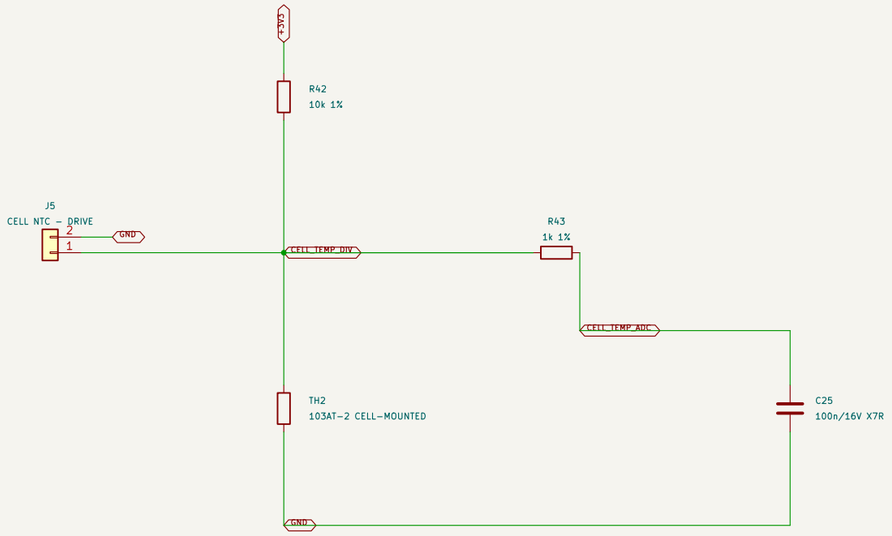
J4 sensor adheres to cell pouch.
Separate from charger and monitor sensors.

Integration review only. Confirm cold insertion and charger fault recovery before lab bring-up.

Existing TH1 and TH2 circuits below remain cell-mounted.





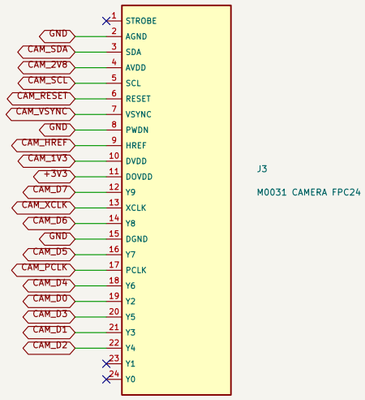


Nate

Sheet: /Battery protection/
File: battery_protection.kicad_sch

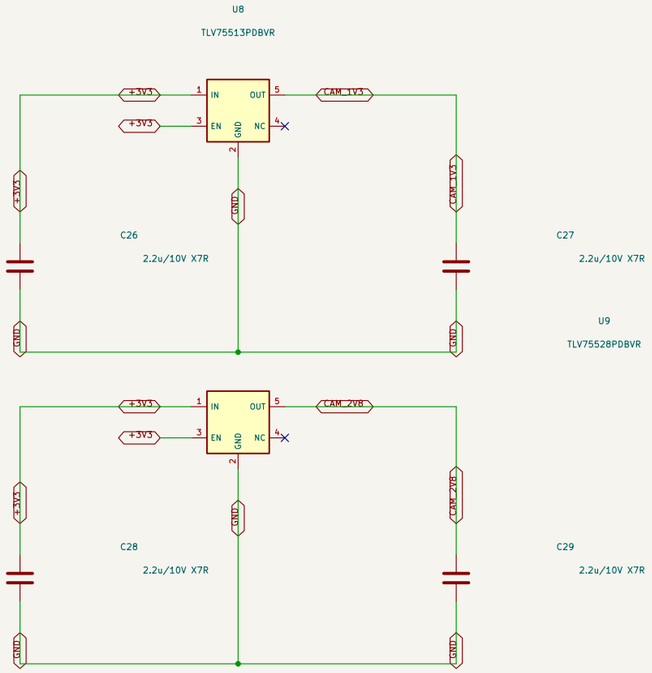
Title: Omni rover | battery protection

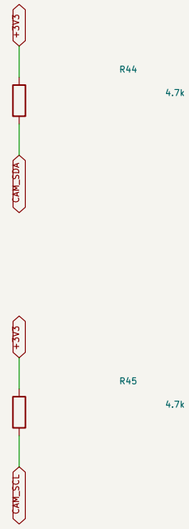
Size: A3	Date:	**Rev: F draft**
KiCad E.D.A. 10.0.8		Id: 7/11



J3

M0031 CAMERA FPC24



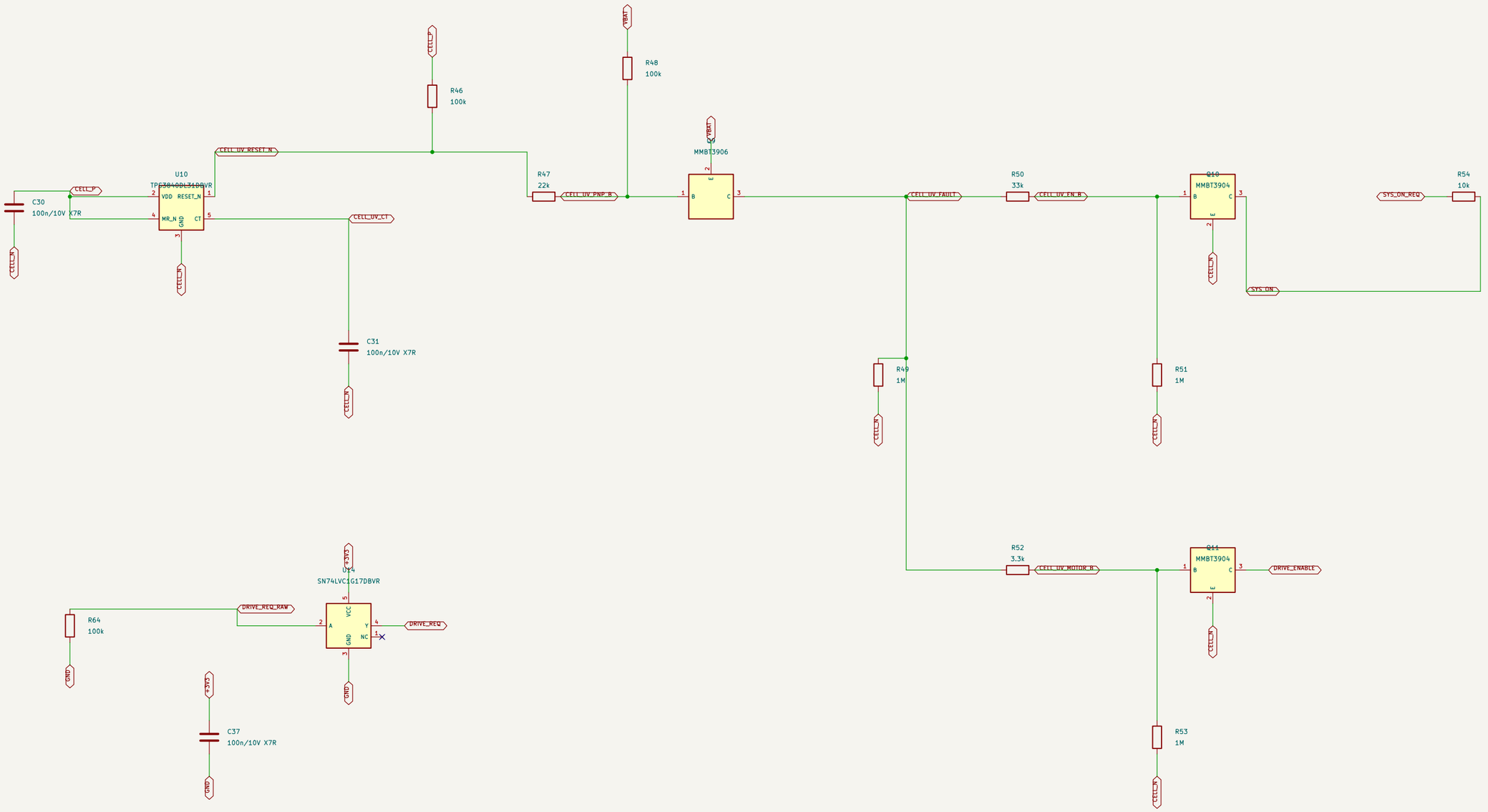


M0031 only. Verify flex contact side, thickness and pin-1 before assembly.
1.3 V core / 2.8 V analog / 3.3 V I-O. No interchangeable generic OV2640 pinout.

Sheet: /Flex camera/	
File: camera.kicad_sch	
Title: OV2640 M0031 flex interface	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: E – prototype**
	Id: 8/11

Independent cell undervoltage shutdown

3.1 V supervisor: load inhibits plus independent discharge cutoff on sheet 11



R46 now pulls RESET_N to raw CELL_P. Q12/Q13 remain effective after protected VBAT disappears.
Q9/Q10/Q11 retain the existing logic and motor inhibits; R64 defaults the motor request low.
U7 retains independent overvoltage/current/short protection and its factory 2.6 V UV threshold.
This is not a zero-leakage storage disconnect: the supervisor, protector and cutoff bias still use cell current.

Nate | Omni camera rover

Sheet: /Cell undervoltage load shutdown/
File: cell_undervoltage.kicad_sch

Title: Independent cell undervoltage shutdown

Size: A3	Date:	**Rev: E – prototype**
KiCad E.D.A. 10.0.8		Id: 9/11

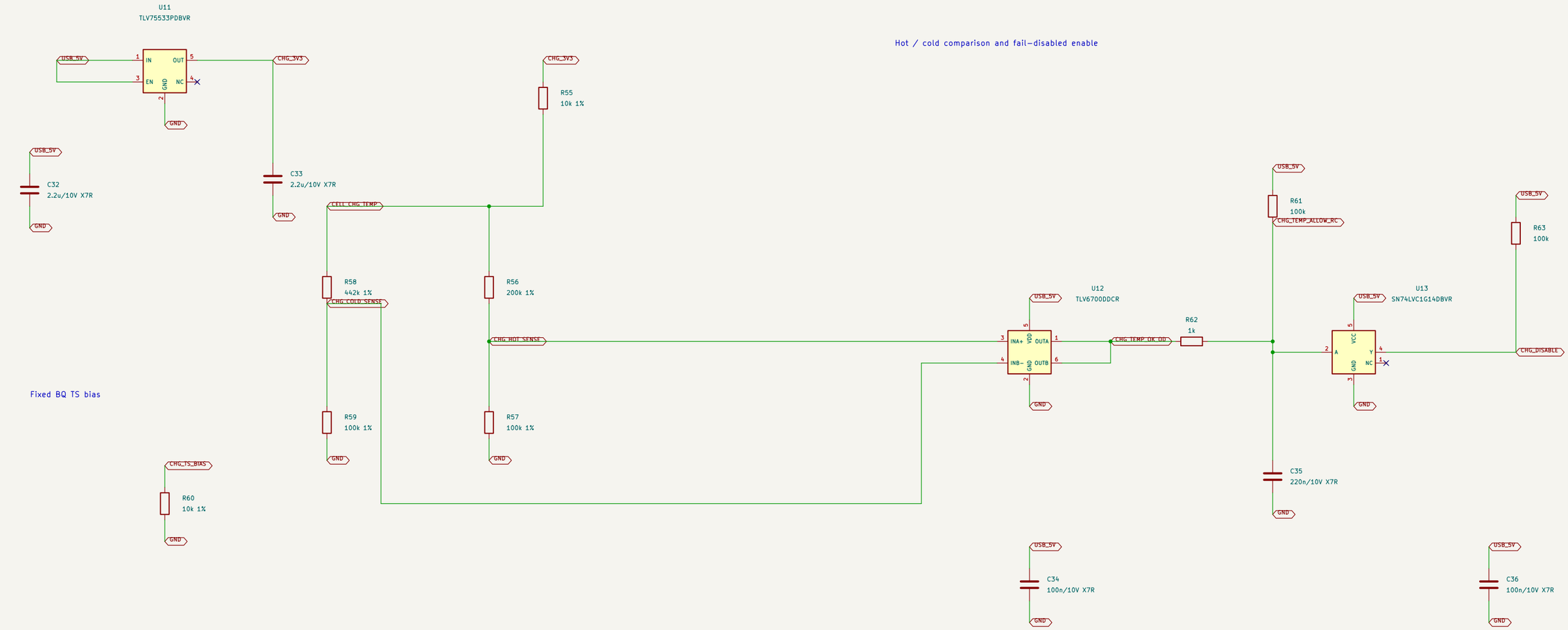
Independent cell temperature charging window

USB-powered reference supply

TH1 at J4: actual cell temperature

Hot / cold comparison and fail-disabled enable

Fixed BQ TS bias



TH1 is an electrically insulated OFF-BOARD 103AT-2, bonded to the cell. Open/short probe disables charging.
U12 open-drain outputs both release only within the temperature window. U13 then drives CE low.
R61/C35 delay charge enable at USB startup; R62 limits capacitor discharge into U12. No ESP32 dependency.
R60 intentionally replaces the direct TS thermistor. This independent circuit now decides cell charge temperature.
Tolerance calculation uses the NTC beta model: verify actual resistance-temperature curve and trip limits on a fixture.

Nate | Omni camera rover

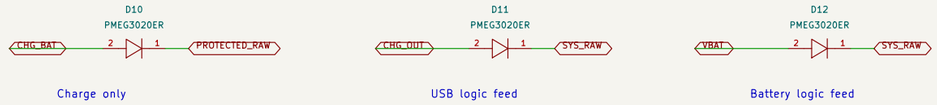
Sheet: /Cell temperature charging window/
File: charge_temperature.kicad_sch

Title: Independent cell temperature charging window

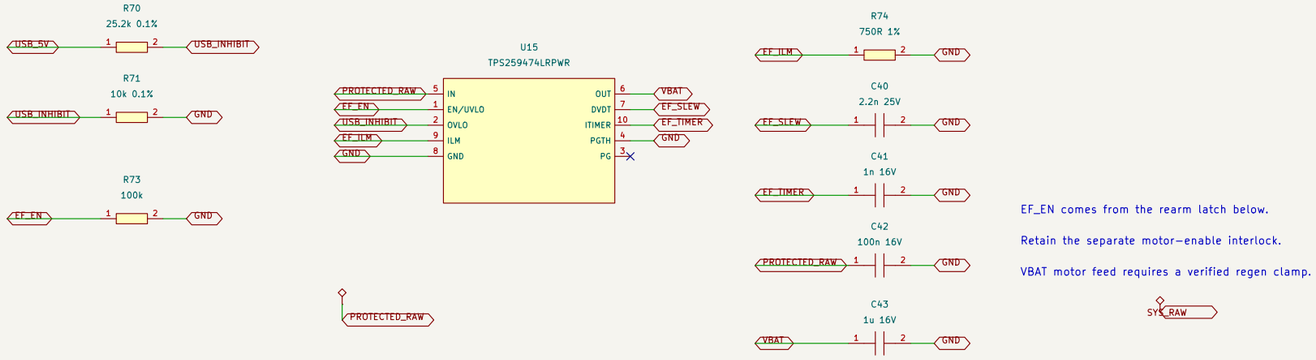
Size: A3	Date:	**Rev: E – prototype**
KiCad E.D.A. 10.0.8		Id: 10/11

1. Charge and logic paths

USB_5V, CHG_BAT and CHG_OUT connect to the Charging sheet. SYS_RAW powers the regulator.



2. Battery discharge switch and USB inhibit



EF_EN comes from the rearm latch below.

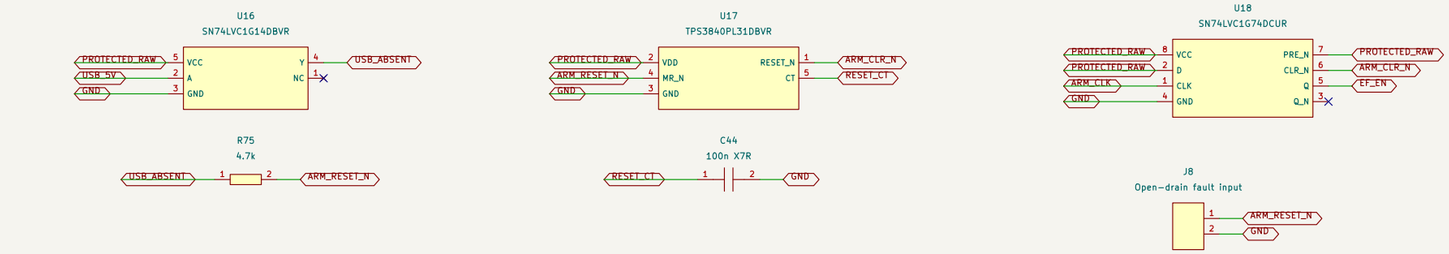
Retain the separate motor-enable interlock.

VBAT motor feed requires a verified regen clamp.

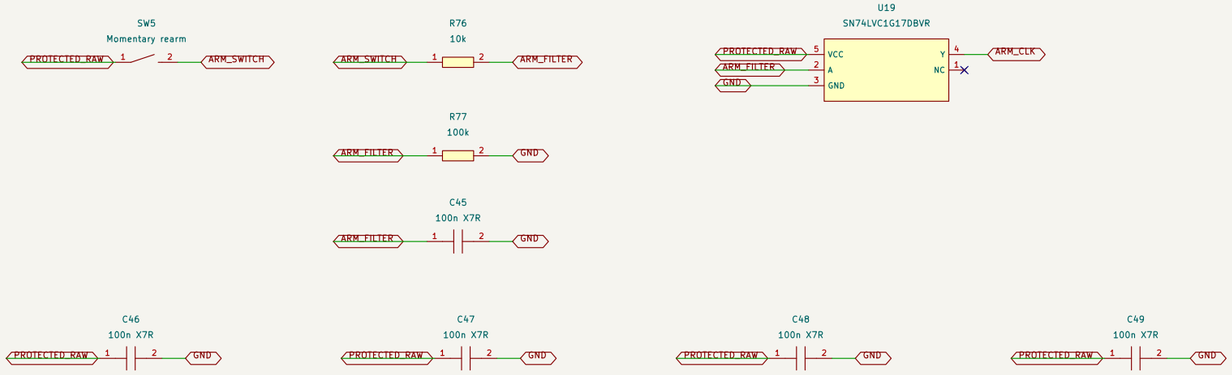
SYS_RAW

3. Hardware rearm after USB, undervoltage or external fault

Wake the protector, then press SW10 after reset releases. A held button has no new rising edge.



J12 is a fault-injection port. The final hard-short detector is external.



C16-C19 are local bypass capacitors for U11-U14 respectively.

Fixture draft. Verify reset timing, held shorts, charge termination and motor transients before routing.

Nate

Sheet: /Rev F USB and drive power path/
File: discharge_cutoff.kicad_sch

Title: Omni rover | USB and drive isolation

Size: A3	Date:	**Rev: F draft**
KiCad E.D.A. 10.0.8		Id: 11/11